FACIAL APPEARANCE GENDER RECOGNITION

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***Abstract* — This project focuses on advancing gender recognition through facial appearance using state-of-the-art deep learning models. The three selected models, ResNet, MobileNet, and EfficientNet, represent cutting-edge architectures in the field. Leveraging the latest versions of each model, our objective is to comprehensively explore their capabilities for accurate gender identification. ResNet, known for its residual learning framework, MobileNet, optimised for mobile and edge devices, and EfficientNet, celebrated for its efficient compound scaling methodology, offer distinct advantages. By shedding light on the suitability of ResNet, MobileNet, and EfficientNet for gender recognition based on facial appearance, this study advances the area. The results are intended to direct future applications in a variety of fields, including marketing analytics, law enforcement, and mental health awareness. The novel aspect of the study is its thorough analysis of three deep learning models, which provides a detailed grasp of each model's advantages and disadvantages when it comes to gender detection from face features.**

Keywords — Gender Recognition, deep learning, CNN, ResNet50, MobileNetV3, EfficientNetV2, facial appearance classification

1. INTRODUCTION

With a significant portion of the tech-savvy population familiarising themselves with cutting-edge technologies, the spotlight on computer vision has undeniably intensified. Deep learning, a subset of machine learning, is gaining prominence in facial recognition applications spanning various industries. A specific facet of facial recognition technology involves discerning an individual's gender based on their facial characteristics.

Gender recognition is a rapidly developing area of computer vision that deals with determining and interpreting an individual's gender from visual data. Conventional techniques include gender labelling photos and videos or manually categorising facial expressions. These methods require a lot of time and are biased. A noteworthy development in recent years has been the increase in popularity of automating gender detection using machine learning algorithms, especially those that use deep learning techniques. These algorithms have proven to perform well in a variety of gender recognition tests, suggesting that they might improve the effectiveness and precision of gender recognition systems.

Facial gender analysis has a wide range of applications, including those in marketing and advertising. Gender analysis is a useful tool that businesses can use to better understand how consumers react to their campaigns or goods, which will help them refine their marketing tactics. Gender recognition technology can also be useful in detecting changes in gender-related signs for possible mental health diagnosis and treatment, as well as in increasing awareness of gender-related concerns. Real-time gender identification in customer service may be used to recognise and cater to gender-specific demands, resulting in better interactions. Law enforcement organisations can also use gender recognition to track and identify shifts in gender-related indications during court cases or police questioning.

Despite advancements, identifying facial gender expression remains challenging due to the intricate and diverse nature of human expressions. Humans convey gender through a combination of facial expressions, body language, and environmental cues, making it a multifaceted task. Culture, mentality, and individual characteristics are some other factors that add to the complexity of gender representation. Constructing an effective and accurate facial gender recognition system necessitates a blend of machine learning methodologies, human expertise, and an understanding of the underlying psychological and social variables influencing gender expression.

The objective of this research is to investigate the viability of several deep learning models, particularly convolutional neural network (CNN) architectures, in tackling the problem of gender recognition based on visual features. There will be suggestions for improving the model's performance, such as modifying the parameters in accordance with the dataset. The efficiency of the model and the accuracy of the predictions made in relation to the data preprocessing and augmentation used on the gender recognition dataset are also closely examined in this work.

2. BACKGROUND / RELATED WORK

2.1 ResNet50

ResNet50 is a deep neural network architecture that is an extended version of the original ResNet. ResNets are known for its effectiveness in training deep neural networks by utilising residual blocks, which include shortcut connections that skip one or more layers. The architecture consists of 50 layers, encompassing both convolutional and fully connected layers. ResNet50 has demonstrated exceptional performance in various image recognition tasks, making it a suitable candidate for gender identification through facial features.

ResNet50 was introduced by He at al. in 2016, revolutionised deep learning by addressing the vanishing gradient problem through residual learning. The idea of residual blocks allows the network to learn residual mappings, making the optimization of extremely deep networks more manageable. The literature underscores the

significance of ResNet50's ability to represent hierarchical facial features, including jawlines, cheekbones, and nuanced expression. Recent studies highlight ResNet50's adaptability to diverse populations, considering age, ethnicity, and cultural influences, addressing challenges in gender recognition.

Kanwarpartap et al. (2023) conducted a pioneering research project that demonstrated the application of the ResNet50 model in face mask detection, specifically tailored for accurately identifying individuals wearing face masks in images or videos, particularly in the context of the COVID-19 pandemic. The researchers employed transfer learning as a fundamental technique, pre-training the ResNet50 model to leverage knowledge from a diverse dataset. To optimise the model's performance, meticulous attention was given to hyperparameter tuning, encompassing adjustments to the number of layers, learning rate, regularisation strength, and the size of the sliding window's stride. This approach resulted in an exceptional accuracy rate of 99% in predicting faces wearing masks, highlighting the efficacy of the ResNet50 model in the critical context of face mask detection.

2.2 MobileNetV3

MobileNetV3 represents the latest evolution in the MobileNet series of lightweight convolutional neural network (CNN) architectures designed for efficient deployment on mobile and edge devices. MobileNetV3 was developed in response to the requirement for neural networks that achieve a balance between computational efficiency and high performance. The architecture is designed to handle mobile devices' problems, such as limited computational resources and power constraints.

MobileNet was introduced by Andrew G. Howard, Menglong Zhu, Bo Chen, Dmitry Kalenichenko, Weijun Wang, Tobias Weyand, Marco Andreetto, and Hartwig Adam at Google Research. They published the original paper titled "MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications" in April 2017. The version 3 of MobileNet, known as MobileNetV3, was also developed by researchers at Google. The primary contributors to MobileNetV3 include Andrew G. Howard, Mark Sandler, Grace Chu, Liang-Chieh Chen, Bo Chen, Mingxing Tan, Weijun Wang, Yukun Zhu, Ruoming Pang, Vijay Vasudevan, Quoc V. Le, and Hartwig Adam. MobileNetV3 was introduced in the paper titled "Searching for MobileNetV3" by the same team in May 2019.

Siying Qian et al. project demonstrates the applicability of MobileNetV3 in real-life scenarios, emphasising its superior balance between efficiency and accuracy on mobile terminals. The motivation behind the development of MobileNetV3 lies in addressing the computational challenges faced by traditional networks on mobile devices. The fusion of techniques like scaling and training-aware neural architecture search (NAS) contributes to its effectiveness.

2.3 EfficientNetV2

EfficientNetV2's motivation is to increase convolutional neural networks' (CNNs) training speed and efficiency while preserving their high parameter efficiency. This version, named after Tan et al. (2021) publication "EfficientNetV2: Smaller Models and Faster Training," expands on the base of the original EfficientNet with the goal of further maximising computing efficiency and training speed.

Two main techniques are combined by EfficientNetV2: scaling and training-aware neural architecture search (NAS). In order to find the best network configurations, neural architecture search entails searching a large architectural space. This procedure is driven by training speed in the case of EfficientNetV2, which results in the selection of designs that promote quicker model convergence during training. A noteworthy inclusion in the architecture is the Fused-MBConv module. When this additional action is incorporated into the search space, the network's efficiency is increased. By carefully using this procedure in the network's early layers, the authors improve the network's capacity to effectively collect pertinent characteristics. The background of EfficientNetV2 is rooted in the quest for refining and accelerating the training process without compromising on the model's overall efficiency and effectiveness.

3. APPROACH

In this study, several CNN architectures will be explored in solving the gender recognition through facial appearances problem. To improve performance, potential changes to these structures and their parameters in relation to the dataset will be suggested. The effects of applying data augmentation and preprocessing to the dataset on model performance and prediction accuracy will be examined. Finding the best deep learning architecture for recognising gender from facial appearances is the aim of this research. We will examine several convolutional neural network (CNN) designs, including ResNet-50, MobileNetV3, and EfficientNetV2, in order to do this. The gender classification face dataset from Kaggle, which is accessible to the public, will be used to train the CNNs. This includes applying strategies like data augmentation and transfer learning. Every model will be trained with identical training and validation splits to provide an equitable comparison. A range of assessment measures will be employed to scrutinize the efficacy of the models. All things considered; this research will offer insightful information about the application of deep learning for gender recognition based on face features.

3.1 ResNet50

ResNet50 is one of the techniques employed in the study of facial appearance gender recognition, representing a robust convolutional neural network architecture. Unlike conventional models, ResNet50 introduces skip connections that facilitate training deep neural networks.

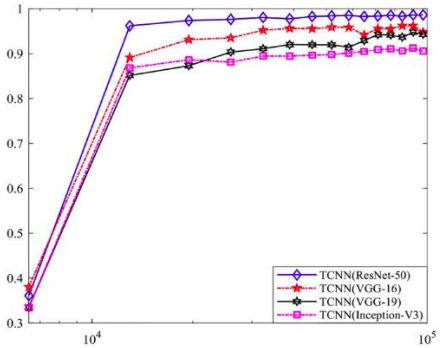


Figure 1: Comparison of ResNet training accuracy

The model's efficiency is highlighted in Figure 1, showcasing its efficiency in training times and accuracy, especially when compared to earlier versions. ResNet50's versatility in capturing complex features from facial images is attributed to its unique residual learning approach. The architecture, which is a stack of residual blocks featuring identity and projection shortcuts, enables the learning of residual functions, enhancing training effectiveness. The sequential integration of global averaging pooling, dense, and dropout layers further refines the model for binary gender classification. ResNet50's adaptability to diverse computer vision task, coupled with its demonstrated success in various domains, positions it as a valuable asset in achieving precise and efficient gender classification based on facial features.

3.2 MobileNetV3

MobileNetv3 is one of the techniques utilised in this study on facial appearance gender recognition. As the most recent model in the MobileNet family, its goal is to outperform other designs in image classification tasks in terms of speed, efficiency, and effectiveness.

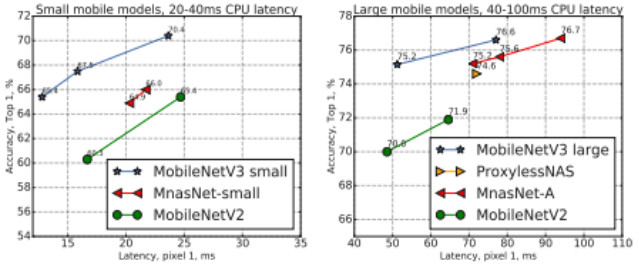


Figure 2: The trade-off between Pixel 1 latency and top-1 ImageNet accuracy

Figure 2 shows the trade-off between Pixel 1 latency and top-1 ImageNet accuracy. It demonstrates that MobileNetV3-Small and MobileNetV3-Large, with multipliers 0.75, 1, and 1.25, showcase an optimal frontier in terms of latency and accuracy. The latency measurements were conducted on a single large core using TFLite. This comparison showcases MobileNetV3 as a next-generation mobile model, indicating its efficiency in terms of inference speed.

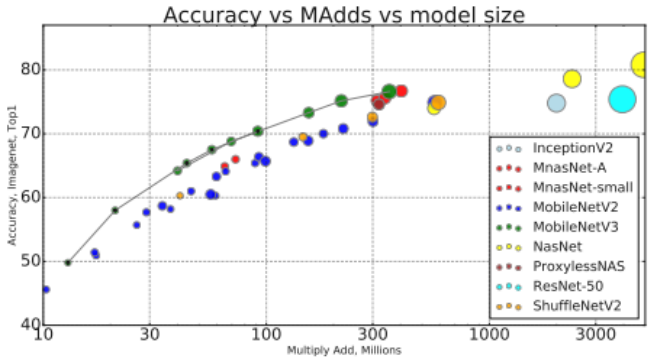


Figure 3: The trade-off between Multiply-Adds (MAdds) and top-1 accuracy

Figure 3 provides insights into the trade-off between Multiply-Adds (MAdds) and top-1 accuracy, allowing a comprehensive comparison of models targeted at different hardware or software frameworks. The figure offers a visual representation of the efficiency and accuracy achieved by MobileNetV3 across different configurations.

Efficiency in MobileNetV3 is achieved through the evolution of efficient building blocks used in earlier models like MobileNetV1, MobileNetV2, and MnasNet. These building blocks, such as depth wise separable convolutions, linear bottlenecks, and inverted residual structures, contribute to computational efficiency while maintaining model performance. MobileNetV3 combines these efficient building blocks, upgrading them with modified swish nonlinearities for improved expressiveness.

To further define MobileNetV3, two models are introduced which are MobileNetV3-Large and MobileNetV3-Small, catering to high and low resource use cases, respectively. These models are crafted through platform-aware Neural Architecture Search (NAS) and NetAdapt, incorporating network improvements to enhance efficiency.

3.3 EfficientNetV2

EfficientNetV2 is one of the techniques utilized in this study on facial appearance gender recognition. As the most recent model in the EfficientNet family, its goal is to outperform other designs in image classification tasks in terms of speed, efficiency, and effectiveness. Figure 4 illustrates how much faster the training times of all the EfficientNetV2 models are compared to the previous EfficientNet version. Additionally, it demonstrates that EfficientNetV2 models have greater accuracy than the previous version.

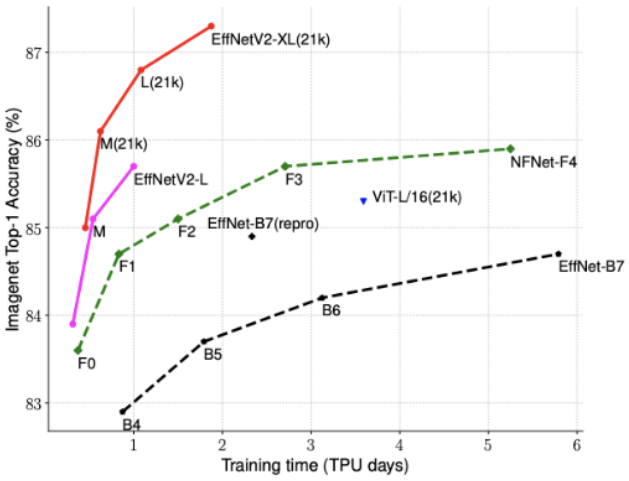


Figure 4: Comparison of EfficientNet training efficiency

EfficientNetV2 differs significantly from its predecessor in terms of architecture. In the early layers, it heavily integrates both the recently announced Fused-MBConv and MBConv (MobileNetV2-like block), with the goal of balancing computational efficiency and model performance. Interestingly, EfficientNetV2 prefers smaller expansion ratios for MBConv since they often result in lower memory access cost. It is clear that smaller 3x3 kernel sizes are preferred, however the model adds extra layers to make up for the lower receptive field that results from this decision.

One notable difference between EfficientNetV2 and the original EfficientNet is the removal of the last stride-1 step. The original stride-1 stage's huge parameter size and memory access cost may have served as the driving force behind this tactical change. EfficientNetV2, with its refined and optimised design that excels in quicker training times and

enhanced parameter efficiency, is a noteworthy breakthrough in the field of efficient convolutional neural networks.

4. EXPERIMENT

4.1 Dataset

In this project, the Gender Classification dataset is used for training, validation, and test of the proposed models. The dataset is publicly available and is retrieved from Kaggle. It was prepared by Gerry (2020). Gerry developed it by utilising the facial photos from the CelebFaces Attributes (CelebA) dataset.

The data consists of 3891 colour images of faces of size 90x120 pixels. The training set consists of 3491 images while the validation and test set contains 200 images each, which will be used for performance metrics. The owner of this dataset developed his own face cropping function, MTCNN to crop the images only including the face in order to distinguish the gender shown in the images.

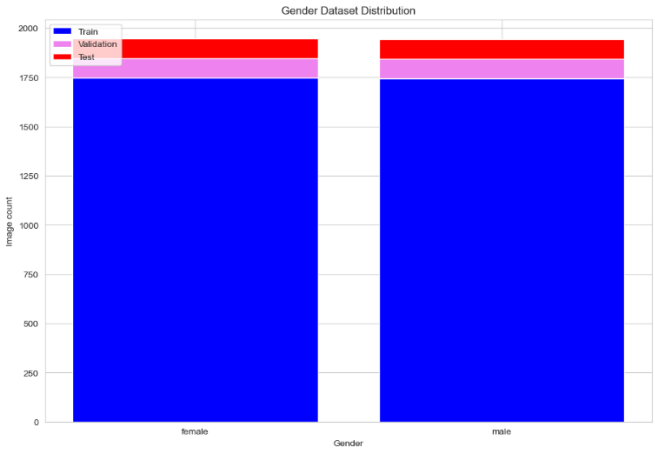


Figure 5: Gender Dataset Distribution

As can be seen from the figure 5 above, even though the dataset is already balanced we applied the augmentation to only the training set as we want to increase our model's generalisation performance by adding more data and diversifying the training dataset. Before the data is fed into the models, ImageDataGenerator is used to do data augmentation for the entire dataset. One useful way to add data to the dataset in Keras is using the ImageDataGenerator class. A collection of data augmentation parameters may be defined and applied automatically to the data as it is fed into the model using this class. This makes it easier to experiment with different data augmentation methods and select the best one for a given use case.

4.2 Evaluation Metrics

This section describes the Accuracy and Confusion Matrix performance evaluation metrics for each model. As a result, the anticipated analysis of the experiment's findings will include charts showing the metrics and the distribution of predictions versus the actual gender classes in addition to a detailed discussion of the metrics listed in the subsections below.

4.2.1 Accuracy

Accuracy is one evaluation metric that is commonly applied to picture classification problems. Accuracy, which is the ratio of correctly predicted picture classes to the total number of photographs, is the most basic performance metric. Classification accuracy is usually expressed as a percentage, with higher values denoting better performance and ease of interpretation.

4.2.2 AUC-ROC Score

AUC-ROC is a metric used to evaluate how effectively a classification model predicts classes. It gauges how well the model can distinguish between the classes and is a useful statistic to use in cases when the classes are not balanced. The true positive rate vs the false positive rate is plotted on the y-axis of the ROC curve, which serves as the foundation for the AUC-ROC. The AUC-ROC stands for area under the ROC curve. A score of 1 indicates perfect discrimination, whereas a score of 0.5 indicates the model performs no better than randomness.

4.2.3 Confusion Matrix

Another often used assessment metric for evaluating a model's performance in image classification tasks is the confusion matrix. It shows the percentage of precise and imprecise predictions the model generated for every class in the dataset. In the confusion matrix, the anticipated class labels are listed on one axis, while the actual class labels are recorded on the other. The number of test set instances that were wrongly allocated to one class when they actually belonged to another is shown by the entries in the matrix. It is easier to spot trends and patterns in the model's performance in this research since the confusion matrix is shown as a heatmap.

4.3 ResNet50

4.3.1 Training History

Figure 6 shows the training and validation accuracy and loss graph for ResNet50. The accuracy on the validation dataset achieves an accuracy of 94.0%. Additionally, the model exhibits exceptional accuracy on the training dataset, reaching an impressive 98.8%. The training and validation loss graph further support the model's convergence, as evidenced by the decreasing trend in both metrics.

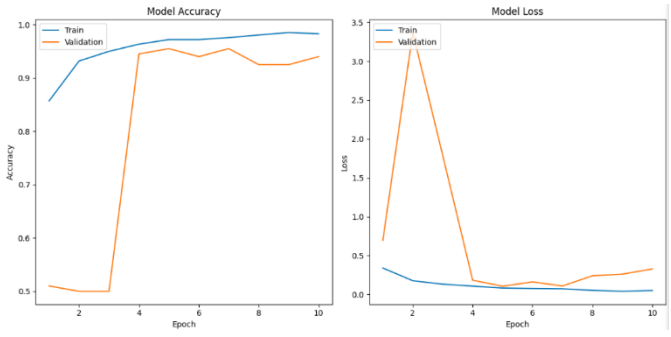


Figure 6: History of Training and Validation graph of ResNet50

The training history plot illustrates the performance of the model throughout the training process across multiple epochs. The model demonstrates a positive trajectory in accuracy, with the training accuracy steadily increasing from approximately 85.7% in the initial epoch to around 98.28% by the 10th epoch. Similarly for the validation accuracy exhibits an upward trend, starting at 51% and reaching approximately 94%. The convergence of training and validation accuracies suggests that the model is effectively

learning and generalising. Regarding loss, the training loss steadily decreased from 0.34 to 0.051, indicating effective fitting to the training data. However, the validation loss initially drops significantly from 0.696 to 0.105 by the 5th epoch but experiences fluctuations thereafter. The gap between training and validation metrics in the early epochs suggest potential overfitting, prompting the need for regularisation techniques. Fine-tuning hyperparameters and incorporating regularisation strategies could further enhance the model's overall performance and generalise better to unseen data.

4.3.2 AUC-ROC and Confusion Matrix

With an outstanding AUC-ROC score of 99.8%, the performance of the ResNet50 model is exceptionally high. This signifies the mode's remarkable ability to accurately distinguish between male and female subjects based on facial appearances. The AUC scores of 1 for both genders, shown in Figure 7. The ROC curve, almost forming a perfect square area, further underscores the mode's precision.

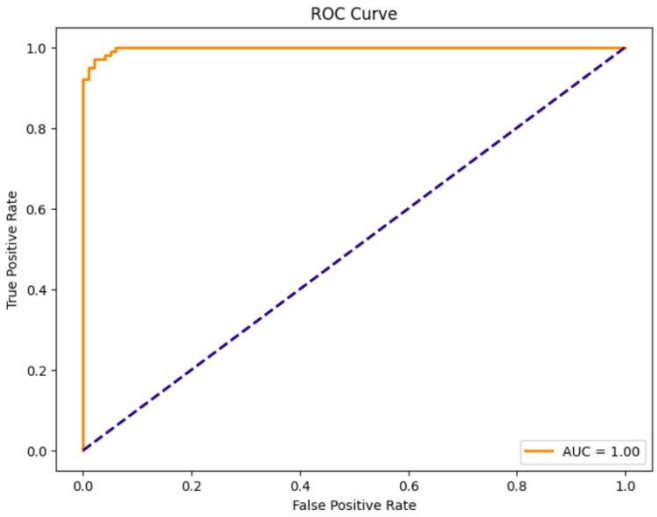


Figure 7: ROC Curve of ResNet50

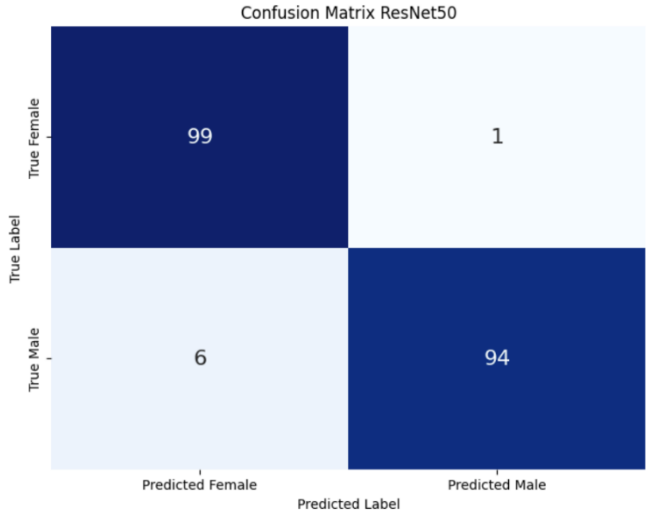


Figure 8: Confusion Matrix for ResNet50

Figure 8 shows the confusion matrix for ResNet50, providing a detailed breakdown of the model's classification performance in gender prediction based on facial appearance. Overall, the model demonstrates strong performance with a high number of correct predictions, particularly for males, and a small number of misclassifications.

4.4 MobileNetV3

4.4.1 Training History

Figure 9 shows the history of training and validation accuracy and loss graph of MobileNetV3. The model has reached a final training accuracy of 99.9% and a validation accuracy of 90.5%.



Figure 9: History of Training and Validation graph of MobileNetV3

First graph represents the history of accuracy. The blue line represents the training accuracy, which starts around 84% and steadily increases to nearly 99% at the end of training (20 epochs). This indicates that the model is effectively learning the patterns in the training data. The orange line represents the validation accuracy, which also increases during training but remains consistently lower than the training accuracy. It starts around 79% and reaches approximately 90.5% at the end. The second graph represents the history of loss. The training loss, which starts around 0.34 and steadily decreases to almost 0.02 by the end of training. This indicates that the model is making increasingly accurate predictions on the training data as it learns. The graphs suggest that the MobileNetv3 model achieved good accuracy on both the training and validation sets.

4.4.2 AUC-ROC and Confusion Matrix

With an AUC-ROC score of 99%, this model performs quite well on the gender recognition task. This model is ideal, as seen by the AUC scores of 1 for both the male and female genders as seen in figure 10 below. The ROC curve is almost a perfect square area, indicating that this model is nearly accurate in all of its predictions.

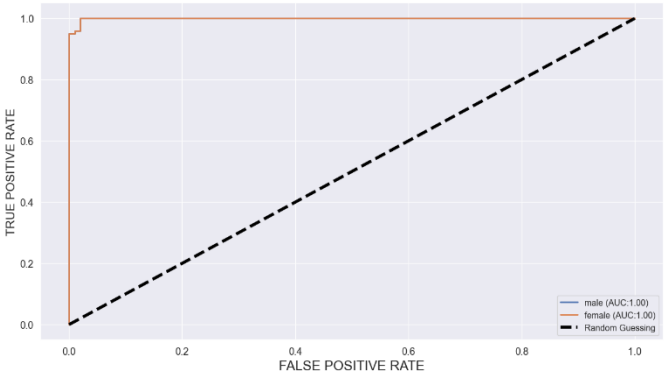


Figure 10: ROC Curve of MobileNetV3

Figure 11 below shows the confusion matrix of MobileNetV3. Both male and female classes are predicted very well as the model only incorrectly predicted 2 instances

of male when it was actually female and 0 for female. The precision for predicting males is 98%, indicating a high proportion of instances predicted as male are indeed male and the precision for predicting females is 100%, also indicating a high proportion of instances predicted as female are indeed female. With a 100% recall rate for female predictions, the model was able to accurately identify 100% of actual female instances.

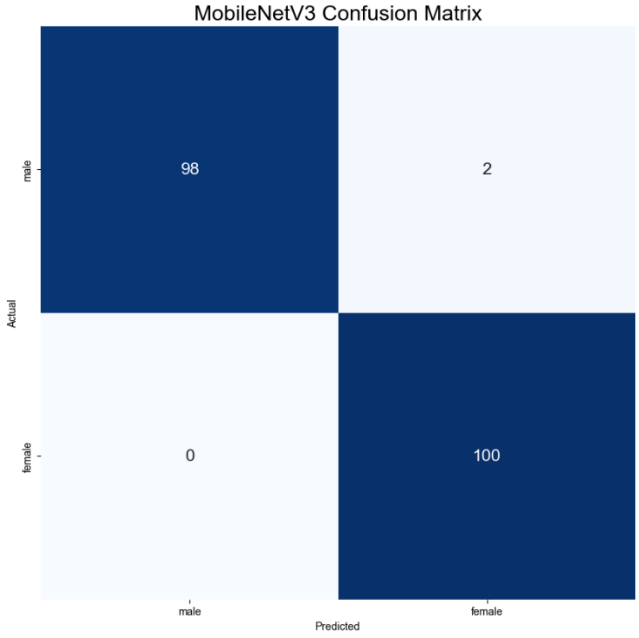


Figure 11: Confusion Matrix for MobileNetV3

Overall, MobileNetV3 showcases impressive gender prediction accuracy, achieving nearly perfect scores for both male and female classifications.

4.5 EfficientNetV2

4.5.1 Training History

Figure 12 shows the history of training and validation accuracy and loss graph of EfficientNetV2. The accuracy of this model on the training dataset is 98.7% and the accuracy on the validation dataset is 87%. The validation loss is greater than the training loss and fluctuates a lot as seen in the figure below.

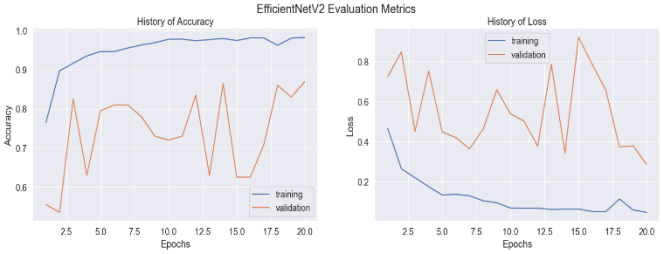


Figure 12: History of Training and Validation graph of EfficientNetV2

This often means that the model is overfitting and is unable to make generalisations on fresh data. To be more precise, the model does well with training data but badly with the newly included data in the validation set. Following the peak at epoch 15, the validation loss decreases until the training is completed. The model may be too complex for the data, or it may have been trained for an extended amount of time, which is a noteworthy explanation for this occurrence. In this case, training can be halted when the loss is low and stable, this is usually known as early stopping. Early stopping is one of the many approaches used to prevent overfitting.

4.5.2 AUC-ROC and Confusion Matrix

With an AUC-ROC score of 97.5%, this model performs quite well on the gender recognition task. This model is ideal, as seen by the AUC scores of 1 for both the male and female genders as seen in figure 13 below. The ROC curve is almost a perfect square area, indicating that this model is nearly accurate in all of its predictions.

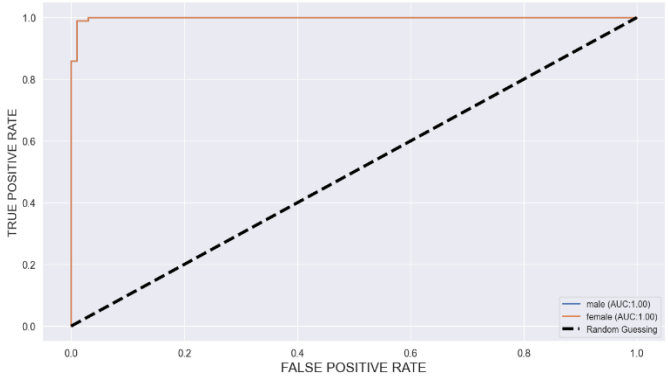


Figure 13: ROC Curve of EfficientNetV2

Figure 14 below shows the confusion matrix of EfficientNetV2. Both male and female classes are predicted very well as the model only incorrectly predicted 1 instance of male when it was actually female and only incorrectly predicted 4 instances of female when it was actually male. The precision for predicting males is 99%, indicating a high proportion of instances predicted as male are indeed male and the precision for predicting females is 96%, also indicating a high proportion of instances predicted as female are indeed female. With a 96% recall rate for male predictions, the model was able to accurately identify 96% of actual male instances.

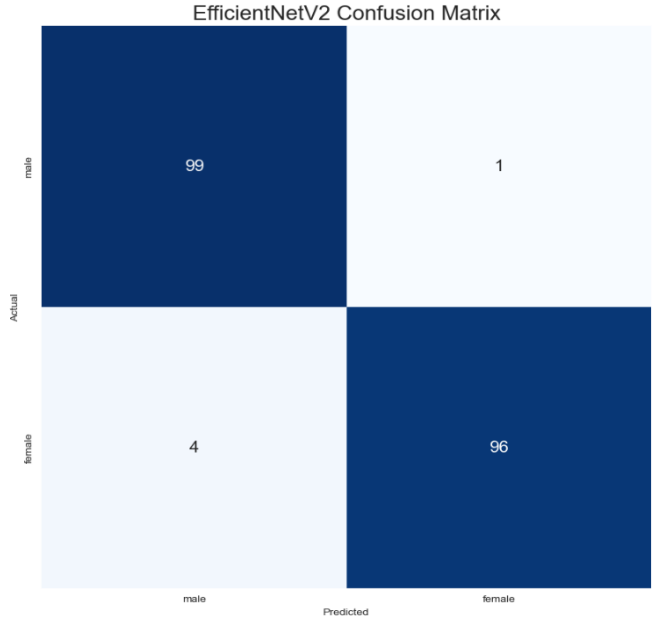


Figure 14: Confusion Matrix for EfficientNetV2

To sum up, EfficientNetV2 demonstrates great gender prediction performance, with both male and female categories showing excellent accuracy and recall. The model's superiority in predicting male gender traits is demonstrated by the fact that it can accurately anticipate more male cases than female ones. The gender prediction model is generally accurate and dependable due to its low rates of false positives and false negatives.

4.6 Overall Analysis

Table 1 shows the side-by-side evaluation of ResNet-50, MobileNetV3, and EfficientNetV2 reveals impressive results in terms of training and test accuracy, as well as AUC-ROC scores. ResNet-50 showcases robust performance with 98.8% training accuracy, 96.5% test accuracy, and an exceptional AUC-ROC score of 99.8%. MobileNetV3 exhibits even higher accuracy rates, boasting 99.9% training accuracy, 99% test accuracy, and a commendable AUC-ROC score of 99%. EfficientNetV2 follows suit with a strong showing, achieving 98.7% training accuracy, 97.5% test accuracy, and a solid AUC-ROC score of 97.5%.

Table 1: Side-by-side evaluation of the models

Model	Train Accuracy	Test Accuracy	AUC-ROC Score
ResNet-50	98.8%	96.5%	99.8%
MobileNetV3	99.9%	99%	99%
EfficientNetV2	98.7%	97.5%	97.5%

Even with the high levels of accuracy, questions concerning stability during training are raised by the variation seen in the validation accuracy and loss graphs for all models. Fluctuations frequently point to possible overfitting and difficulties with generalisation. Nonetheless, the constant high test accuracy indicates that these models function well on unseen data, mitigating concerns about overfitting. All of the models' remarkably high AUC-ROC scores attest to their effectiveness in differentiating between classes and identifying subtle patterns in the data. This has great promise for practical uses where accurate classification is essential.

Considering the overall performance metrics and acknowledging the fluctuation in validation metrics during training, these models demonstrate a remarkable potential for real-world applications. Considering the overall performance metrics and acknowledging the fluctuation in validation metrics during training, these models demonstrate a remarkable potential for real-world applications. Their high accuracy rates and strong AUC-ROC scores indicate a capacity to generalise well to new, unseen data. To further enhance their reliability, fine-tuning hyperparameters, implementing early stopping mechanisms, or exploring additional regularisation techniques may be considered. With appropriate considerations for potential overfitting, these models hold promise for deployment in practical scenarios, ranging from image recognition systems to real-time applications in industries requiring accurate and efficient classification. Figure 15, 16 and 17 below shows a sample of predictions using every proposed model.



Figure 15: Model Prediction visualisation of ResNet50



Figure 16: Model Prediction visualisation of MobileNetV3



Figure 17: Model Prediction visualisation of EfficientNetV2

5. CONCLUSION

In conclusion, the comparative evaluation of ResNet-50, MobileNetV3, and EfficientNetV2 underscores their commendable performance in gender recognition. These models exhibit high training and test accuracies, with ResNet-50 achieving 98.8% and 96.5%, MobileNetV3 showcasing 99.9% and 99%, and EfficientNetV2 delivering 98.7% and 97.5% respectively. The robustness of these models is further emphasised by their exceptional AUC-ROC scores, signifying their proficiency in distinguishing between genders. The models' capacity to successfully generalise to unknown data is indicated by their continuously high test accuracy, even when the validation accuracy and loss graphs show variations throughout training. This shows that the models effectively address overfitting issues and produce accurate predictions in practical settings.

Every model has a distinct set of advantages given the complicated trade-offs between model complexity, computing efficiency, and accuracy. MobileNetV3 performs exceptionally well at high accuracy rates, ResNet-50 has a strong AUC-ROC score, and EfficientNetV2 balances efficiency and accuracy. The selection of a model is contingent upon particular application needs and resource concerns.

In the realm of gender recognition, these models present promising avenues for deployment in diverse fields, including marketing analytics, mental health awareness, and law enforcement. Continuous improvement and optimisation of these models, together with fairness and interpretability issues, will help ensure their smooth integration into practical applications as technology develops. ResNet-50, MobileNetV3, and EfficientNetV2 have all demonstrated success together, establishing them as useful instruments in the field of gender recognition and opening doors for further developments in computer vision applications.

6. REFERENCES

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